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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	07/01/2025	Maximum Marks	10 + 50
Course Code	HS351TA	Duration	120 Min
Sem	V	CIE -II	
ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY RIGHTS			

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
Part – A				
1	_____ component is key in assessing market feasibility	1	1	2
2	What does break-even analysis determine?	1	1	3
3	Which strategy in Porter's generic strategies focuses on offering unique products?	1	1	3
4	What does a proof of concept primarily demonstrate?	1	1	2
5	Why is prototyping important in the ideation process?	2	1	2
6	In order to register Industrial Design, Design should be new and _____	1	1	4
7	Can a colour by itself be subject matter of registration of industrial design? If yes, justify your statement.	2	2	4
8	Upon expiry of registration of a design under Designs Act 2000 it becomes _____ property	1	1	4
Part –B				
1	A medium-sized technology company in India develops an innovative algorithm for optimizing supply chain logistics. To maintain its competitive edge, the company decides not to patent the algorithm but to treat it as a trade secret. Over time, a former employee joins a competitor and allegedly discloses key aspects of the algorithm. The company must navigate the legal landscape to protect its trade secrets. Questions: i. What legal tools and mechanisms are available in India to protect trade secrets? ii. Analyze the company's decision to treat the algorithm as a trade secret instead of patenting it. What are the advantages and risks of this approach? What preventive measures could the company have implemented to minimize the risk of trade secret leakage?	10	3	4



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Academic year 2024-2025 (ODD Sem)

2	A furniture design firm creates a unique chair design that combines aesthetics with ergonomic principles. The design becomes a market hit, but a competitor begins manufacturing and selling a chair with an almost identical appearance. The firm initiates legal proceedings to obtain design protection and stop the competitor's activities. Questions: i. What features of the chair design qualify it for industrial design protection?	10	3	4
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Academic year 2024-2025 (ODD Sem)**DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT**

Date	07 January 2025	Maximum Marks	10 + 50
Course Code	IM352IA	Duration	120 Min
Sem	V	CIE – II	
OPERATIONS MANAGEMENT			

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
Part – A				
1	In services, the process structure is primarily influenced by the level of _____ and customization.	1	1	2
2	In manufacturing, a _____ process is ideal for high-volume, standardized production.	1	2	2
3	The first step in a systematic approach to long-term capacity decisions is _____ future demand.	1	2	2
4	A systematic approach to capacity planning involves forecasting, identifying alternatives, and evaluating _____.	1	1	2
5	The _____ component of a forecast captures recurring patterns over specific periods, such as seasons or months.	1	2	3
6	The smoothing constant in exponential smoothing ranges between 0 and _____.	1	1	3
7	_____ provides a detailed plan of what needs to be produced and when, serving as an input for MRP.	1	1	3
8	The _____ file contains information about the quantities of materials required to make a product.	1	2	3
9	_____ integrates all business processes, including finance, HR, and supply chain, into a unified system.	1	2	3
10	_____ ensures that operational capabilities are in line with the company's strategic goals.	1	2	3
Part –B				
1	Explain the major process strategy decisions and their implications for operations.	10	2	2



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Academic year 2024-2025 (ODD Sem)

2	Richard Garber is the head designer for Matthews and Novak Design Company. Garber has been called in to design the layout for a newly constructed office building. From statistical samplings over the past three months, Garber developed the following closeness matrix for daily trips between the department’s offices. a. If other factors are equal, which two offices should be located closest together? b. Figure shows an alternative layout for the department. What is the total weighted-distance score for this plan based on rectilinear distance and assuming that offices A and B are three units of distance apart? CLOSENESS MATRIX <table><tr><td></td><td colspan="6">Trips Between Departments</td></tr><tr><td>Department</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td></tr><tr><td>A</td><td>—</td><td>25</td><td>90</td><td></td><td>105</td><td>185</td></tr><tr><td>B</td><td></td><td>—</td><td></td><td></td><td>125</td><td>125</td></tr><tr><td>C</td><td></td><td></td><td>—</td><td></td><td>25</td><td></td></tr><tr><td>D</td><td></td><td></td><td></td><td>—</td><td></td><td>105</td></tr><tr><td>E</td><td></td><td></td><td></td><td></td><td>—</td><td></td></tr><tr><td>F</td><td></td><td></td><td></td><td></td><td></td><td>—</td></tr></table> <table><tr><td>C</td><td>F</td><td>A</td></tr><tr><td>B</td><td>E</td><td>D</td></tr></table>		Trips Between Departments						Department	A	B	C	D	E	F	A	—	25	90		105	185	B		—			125	125	C			—		25		D				—		105	E					—		F						—	C	F	A	B	E	D	10	3	2
	Trips Between Departments																																																																	
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C			—		25																																																													
D				—		105																																																												
E					—																																																													
F						—																																																												
C	F	A																																																																
B	E	D																																																																
3	The monthly demand for units manufactured by the Acme Rocket Company has been as follows: <table><tr><td>Month</td><td>Units</td><td>Month</td><td>Units</td></tr><tr><td>May</td><td>100</td><td>September</td><td>105</td></tr><tr><td>June</td><td>80</td><td>October</td><td>110</td></tr><tr><td>July</td><td>110</td><td>November</td><td>125</td></tr><tr><td>August</td><td>115</td><td>December</td><td>120</td></tr></table> i. Use the exponential smoothing method to forecast the number of units for June to January. The initial forecast for May was 105 units; $\alpha = 0.2$. ii. Calculate the absolute percentage error for each month from June through December and the MAD and MAPE of forecast error as of the end of December. iii. Calculate the tracking signal as of the end of December. What can you say about the performance of your forecasting method?	Month	Units	Month	Units	May	100	September	105	June	80	October	110	July	110	November	125	August	115	December	120	10	3	3																																										
Month	Units	Month	Units																																																															
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August	115	December	120																																																															



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Academic year 2024-2025 (ODD Sem)

4	Refer to the bill of materials for product A. If there is no existing inventory and no scheduled receipts, how many units of items G, E, and D must be purchased to produce five units of end item A?	10	4	3
5	Explain how the concept of dependent demand in material requirements planning is fundamental to resource planning.	10	2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution		Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
		Quiz	—	4	6	—	4	6	—	—	—	—
		Test	—	20	30	—	—	20	20	10	—	—



Academic year 2024-2025 (ODD Sem)

**DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT**

Date	7th January 2025	Maximum Marks	50
Course Code	IM353IA	Duration	90 Min
Sem	V	CIE – II	
QUALITY ASSURANCE			

Note:

1. Answer all the Questions.

PART – A

Max Marks: 10

Sl. No	Questions	M	BT	CO
1.	Differentiate between Attribute control chart and Variable control chart.	2	L2	CO1
2.	What does an R-chart measure, and why is it important in statistical process control?	2	L1	CO2
3.	What is the key difference between a p-chart and a np-chart?	2	L2	CO3
4.	Expand the acronyms: AQL, UTPD	2	L2	CO3
5.	Write two applications of acceptance sampling.	2	L2	CO2

PART – B

Sl. No	Questions	M	BT	CO																																																							
1.	A TiW layer is deposited on a substrate using a sputtering tool. The following table contains layer thickness measurements (in Angstroms) on 10 subgroups of four substrates. Set up $\bar{X}$ and s charts on this process and compute trial control limits. <table border="1"> <thead> <tr> <th>Subgroup</th><th>x_1</th><th>x_2</th><th>x_3</th><th>x_4</th></tr> </thead> <tbody> <tr><td>1</td><td>459</td><td>449</td><td>435</td><td>450</td></tr> <tr><td>2</td><td>443</td><td>440</td><td>442</td><td>442</td></tr> <tr><td>3</td><td>457</td><td>444</td><td>449</td><td>444</td></tr> <tr><td>4</td><td>469</td><td>463</td><td>453</td><td>438</td></tr> <tr><td>5</td><td>443</td><td>457</td><td>445</td><td>454</td></tr> <tr><td>6</td><td>444</td><td>456</td><td>456</td><td>457</td></tr> <tr><td>7</td><td>445</td><td>449</td><td>450</td><td>445</td></tr> <tr><td>8</td><td>446</td><td>455</td><td>449</td><td>452</td></tr> <tr><td>9</td><td>444</td><td>452</td><td>457</td><td>440</td></tr> <tr><td>10</td><td>432</td><td>463</td><td>463</td><td>443</td></tr> </tbody> </table>	Subgroup	x_1	x_2	x_3	x_4	1	459	449	435	450	2	443	440	442	442	3	457	444	449	444	4	469	463	453	438	5	443	457	445	454	6	444	456	456	457	7	445	449	450	445	8	446	455	449	452	9	444	452	457	440	10	432	463	463	443	10	3	CO3
Subgroup	x_1	x_2	x_3	x_4																																																							
1	459	449	435	450																																																							
2	443	440	442	442																																																							
3	457	444	449	444																																																							
4	469	463	453	438																																																							
5	443	457	445	454																																																							
6	444	456	456	457																																																							
7	445	449	450	445																																																							
8	446	455	449	452																																																							
9	444	452	457	440																																																							
10	432	463	463	443																																																							
2.	Write a note on Standardized control chart. What is the process capability? Explain any one of the techniques of process capability analysis.	10	2	CO1																																																							



Academic year 2024-2025 (ODD Sem)

3.	Lots of Bushing screws from a single supplier were inspected 100 percent with the following results. <table><tr><td>Lot No.</td><td>Amount in lot</td><td>Number rejected</td></tr><tr><td>1</td><td>6000</td><td>183</td></tr><tr><td>2</td><td>6000</td><td>131</td></tr><tr><td>3</td><td>10000</td><td>184</td></tr><tr><td>4</td><td>1783</td><td>23</td></tr><tr><td>5</td><td>3089</td><td>171</td></tr><tr><td>6</td><td>5774</td><td>65</td></tr><tr><td>7</td><td>3000</td><td>46</td></tr><tr><td>8</td><td>2724</td><td>85</td></tr></table> Draw an appropriate control chart for the data. What can you conclude on the state of control?	Lot No.	Amount in lot	Number rejected	1	6000	183	2	6000	131	3	10000	184	4	1783	23	5	3089	171	6	5774	65	7	3000	46	8	2724	85	10	2	CO2
Lot No.	Amount in lot	Number rejected																													
1	6000	183																													
2	6000	131																													
3	10000	184																													
4	1783	23																													
5	3089	171																													
6	5774	65																													
7	3000	46																													
8	2724	85																													
4.	Draw the OC Curve, AOQ Curve, ASN Curve and ATI Curve for the single sampling plan with $N = 5000$, $n = 100$ and $c = 2$	10	4	CO4																											
5.	A factory inspects different samples of a product, recording the number of defects in each sample. The sample size is constant, and the number of defects is provided below: <table><tr><td>Sample No.</td><td>Number of Defects</td></tr><tr><td>1</td><td>5</td></tr><tr><td>2</td><td>8</td></tr><tr><td>3</td><td>6</td></tr><tr><td>4</td><td>4</td></tr><tr><td>5</td><td>7</td></tr><tr><td>6</td><td>9</td></tr><tr><td>7</td><td>5</td></tr><tr><td>8</td><td>6</td></tr></table> i. Draw an appropriate control chart for the data. Evaluate the process. ii. Plot the centreline, upper control limit (UCL), and lower control limit (LCL).	Sample No.	Number of Defects	1	5	2	8	3	6	4	4	5	7	6	9	7	5	8	6	10	2	CO2									
Sample No.	Number of Defects																														
1	5																														
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8	6																														

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4
	Quiz	Max Marks	02	04	04	–	2	8	–	–
	Test		10	20	10	10	–	30	10	10



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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	8 th January 2025	Maximum Marks	10 + 50
Course Code	IM254TA	Duration	120 Mins
Sem	V Semester	CIE – II	
FINANCIAL ACCOUNTING AND COSTING			

Note:

1. Answer all the Questions.

PART – A

Sl. No	Questions	M	BT	CO
1	What is a Trial Balance, and why is it prepared?	02	2	CO2
2	How is Gross Profit calculated?	02	1	CO3
3	Why is depreciation included in the Income Statement?	02	2	CO3
4	What is the basic accounting equation represented in a Balance Sheet?	02	1	CO3
5	What is the difference between current and non-current assets?	02	1	CO1

PART – B

Sl. No	Questions	M	BT	CO		
1.	The following are the ledger balances of a business as of December 31, 2024. Prepare a trail balance.	10	3	CO2		
	Account				Amount (₹)	Type
	Cash				50,000	Debit
	Capital				1,50,000	Credit
	Sales				2,00,000	Credit
	Purchases				1,00,000	Debit
	Furniture				25,000	Debit
	Salaries				30,000	Debit
	Accounts Receivable				45,000	Debit
Accounts Payable	40,000	Credit				



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Academic year 2024-2025 (ODD Sem)

2.	Prepare a Profit and Loss Account for Green Enterprises for the year ending December 31, 2024, from the following details:	10	4	CO3																								
	<table><tr><th>Particulars</th><th>Amount (₹)</th></tr><tr><td>Sales Revenue</td><td>7,50,000</td></tr><tr><td>Purchases</td><td>4,00,000</td></tr><tr><td>Opening Inventory</td><td>1,20,000</td></tr><tr><td>Closing Inventory</td><td>80,000</td></tr><tr><td>Wages</td><td>50,000</td></tr><tr><td>Office Expenses</td><td>40,000</td></tr><tr><td>Utilities</td><td>30,000</td></tr><tr><td>Depreciation on Machinery</td><td>20,000</td></tr></table>	Particulars	Amount (₹)	Sales Revenue	7,50,000	Purchases	4,00,000	Opening Inventory	1,20,000	Closing Inventory	80,000	Wages	50,000	Office Expenses	40,000	Utilities	30,000	Depreciation on Machinery	20,000									
Particulars	Amount (₹)																											
Sales Revenue	7,50,000																											
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Wages	50,000																											
Office Expenses	40,000																											
Utilities	30,000																											
Depreciation on Machinery	20,000																											
3.	Prepare a Balance Sheet for Green Valley Enterprises as of December 31, 2024, from the following details:	10	3	CO3																								
	<table><tr><th>Particulars</th><th>Amount (₹)</th><th>Particulars</th><th>Amount (₹)</th></tr><tr><td>Capital</td><td>5,00,000</td><td>Cash</td><td>1,00,000</td></tr><tr><td>Bank Loan</td><td>3,00,000</td><td>Building</td><td>5,00,000</td></tr><tr><td>Accounts Payable</td><td>1,50,000</td><td>Inventory</td><td>2,00,000</td></tr><tr><td>Salaries Payable</td><td>50,000</td><td>Furniture</td><td>80,000</td></tr><tr><td>Accounts Receivable</td><td>1,20,000</td><td>Equipment</td><td>1,50,000</td></tr></table>	Particulars	Amount (₹)	Particulars	Amount (₹)	Capital	5,00,000	Cash	1,00,000	Bank Loan	3,00,000	Building	5,00,000	Accounts Payable	1,50,000	Inventory	2,00,000	Salaries Payable	50,000	Furniture	80,000	Accounts Receivable	1,20,000	Equipment	1,50,000			
Particulars	Amount (₹)	Particulars	Amount (₹)																									
Capital	5,00,000	Cash	1,00,000																									
Bank Loan	3,00,000	Building	5,00,000																									
Accounts Payable	1,50,000	Inventory	2,00,000																									
Salaries Payable	50,000	Furniture	80,000																									
Accounts Receivable	1,20,000	Equipment	1,50,000																									
4.	From the following information, prepare a cost sheet for XYZ Manufacturing Co. for the year ending March 31, 2025:	10	3	CO4																								
	<table><tr><th>Particulars</th><th>Amount (₹)</th></tr><tr><td>Raw Materials Consumed</td><td>3,00,000</td></tr><tr><td>Direct Labor</td><td>2,00,000</td></tr><tr><td>Factory Overheads</td><td>80,000</td></tr><tr><td>Office and Administrative Overheads</td><td>40,000</td></tr><tr><td>Selling and Distribution Overheads</td><td>30,000</td></tr><tr><td>Opening Stock of Finished Goods</td><td>50,000</td></tr><tr><td>Closing Stock of Finished Goods</td><td>60,000</td></tr><tr><td>Profit Percentage on Cost</td><td>25%</td></tr></table>	Particulars	Amount (₹)	Raw Materials Consumed	3,00,000	Direct Labor	2,00,000	Factory Overheads	80,000	Office and Administrative Overheads	40,000	Selling and Distribution Overheads	30,000	Opening Stock of Finished Goods	50,000	Closing Stock of Finished Goods	60,000	Profit Percentage on Cost	25%									
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Closing Stock of Finished Goods	60,000																											
Profit Percentage on Cost	25%																											
5.	Explain the three main elements of cost sheet with examples. How are they classified and used in cost accounting?	10	3	CO4																								

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	02	02	06	—	—	06	04	—	—	—	—
		Test	—	10	20	20	—	—	—	30	20	—	—

**Academic year 2024-2025 (ODD Sem)****DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT**

Date	9th January 2025	Maximum Marks	50
Course Code	IM355TBD	Duration	90 Min
Sem	V	CIE – 2	
ADVANCED DECISION MODELLING (Professional Core Elective)			

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO																											
Part A																															
1.	Mention the situations that make the replacement of items necessary?	02	2	1																											
2.	What are the three strategies of replacement of items which follow sudden failure mechanisms?	02	2	3																											
3.	State two differences between preventive maintenance and breakdown maintenance.	02	2	2																											
4.	Give one advantage and one disadvantage of individual replacement policies.	02	2	2																											
5.	State the main objectives of using the EOQ model?	02	1	3																											
Part B																															
1	A firm is considering the replacement of a machine, whose cost price is Rs 12,200, and its scrap value is Rs 200. From experience the running (maintenance and operating) costs are found to be as follows:<table><tr><td>Year:</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>Running Cost (Rs):</td><td>200</td><td>500</td><td>800</td><td>1200</td><td>1800</td><td>2500</td><td>3200</td><td>4000</td></tr></table>When should the machine be replaced?	Year:	1	2	3	4	5	6	7	8	Running Cost (Rs):	200	500	800	1200	1800	2500	3200	4000	10	3	1									
Year:	1	2	3	4	5	6	7	8																							
Running Cost (Rs):	200	500	800	1200	1800	2500	3200	4000																							
2	The data on the running costs per year and resale price of equipment A, whose purchase price is Rs 2,00,000 are as follows:<table><tr><td>Year:</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td></td></tr><tr><td>Running cost (Rs):</td><td>30,000</td><td>38,000</td><td>46,000</td><td>58,000</td><td>72,000</td><td>90,000</td><td>1,10,000</td><td></td></tr><tr><td>Resale value (Rs):</td><td>1,00,000</td><td>50,000</td><td>25,000</td><td>12,000</td><td>8,000</td><td>8,000</td><td>8,000</td><td></td></tr></table> i. What is the optimum period of replacement? ii. When equipment A is two years old, equipment B, which is a new model for the same usage, is available. The optimum period for replacement is 4 years with an average cost of Rs 72,000. Should equipment A be changed with equipment B? If so, when?	Year:	1	2	3	4	5	6	7		Running cost (Rs):	30,000	38,000	46,000	58,000	72,000	90,000	1,10,000		Resale value (Rs):	1,00,000	50,000	25,000	12,000	8,000	8,000	8,000		10	3	2
Year:	1	2	3	4	5	6	7																								
Running cost (Rs):	30,000	38,000	46,000	58,000	72,000	90,000	1,10,000																								
Resale value (Rs):	1,00,000	50,000	25,000	12,000	8,000	8,000	8,000																								

**Academic year 2024-2025 (ODD Sem)**

3	A company is considering the purchase of a new machine at Rs 15,000. The economic life of the machine is expected to be 8 years. The salvage value of the machine at the end of the life will be Rs 3,000. The annual running cost is estimated to be Rs 7,000. i. Assuming an interest rate of 5 per cent, determine the present worth of future costs of the proposed machine. ii. Compare the new machine with the presently-owned machine that has an annual operating cost of Rs 5,000 and cost of repair Rs 1,500 in the second year, with an annual increase of Rs 500 in the subsequent years of its life.	10	3	3																		
4	The following failure rates have been observed for a certain type of bulbs. <table border="1"><tr><td>End of Week:</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>Probability of failure to date:</td><td>0.05</td><td>0.13</td><td>0.25</td><td>0.43</td><td>0.68</td><td>0.88</td><td>0.96</td><td>1.0</td></tr></table> The cost of replacing an individual bulb is Rs. 2.25. if the cost of group replacement is 60 paise per bulb and the total number of bulbs is 1000. What is best interval between group replacements?	End of Week:	1	2	3	4	5	6	7	8	Probability of failure to date:	0.05	0.13	0.25	0.43	0.68	0.88	0.96	1.0	10	3	3
End of Week:	1	2	3	4	5	6	7	8														
Probability of failure to date:	0.05	0.13	0.25	0.43	0.68	0.88	0.96	1.0														
5	Differentiate between Continuous Review and Periodic Review Inventory Systems by the following aspects: i. Monitoring ii. Reorder Trigger iii. Stockouts Risks iv. System Complexity v. Suitability	10	2	2																		

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	1	2	2	—	1	2	2	—	—	—
	Test	Max Marks	10	20	20	—	—	10	40	—	—	—
